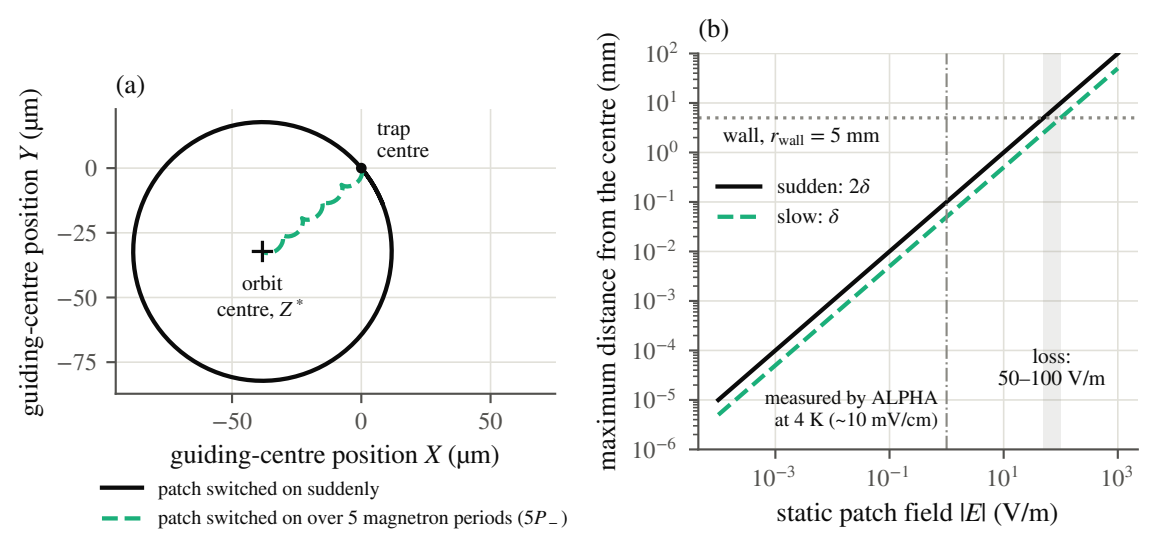
(a)
trap
centre
orbit
centre, Z^*
guiding-centre position Y (μm)
guiding-centre position X (μm)
0
−25
−50
−75
−50
0
50
patch switched on suddenly
patch switched on over 5 magnetron periods ($5P_-$)
(b)
wall, $r_{\text{wall}} = 5$ mm
sudden: 2δ
slow: δ
measured by ALPHA
at 4 K (~10 mV/cm)
loss:
50–100 V/m
maximum distance from the centre (mm)
static patch field $|E|$ (V/m)
10^{2}
10^{1}
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
10^{-3}
10^{-1}
10^{1}
10^{3}